Assessing Spoken Discourse in Aphasia Using Multimodal Artificial Intelligence

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Background

Main concept analysis (MCA) provides criterion-based assessment of discourse informativeness in aphasia, demonstrating sensitivity to severity and ecological validity. However, MCA adoption remains limited by:

- Labor-intensive manual development of concept inventories requiring healthy control norming
- Reliance on static stimuli (e.g., Cookie Theft picture) lacking ecological validity
- Inability to capture dynamic processing demands of real-world communication

Recent advances in large language models (LLMs) enable automated, multimodal MCI generation:

- Multimodal LLMs can process video, audio, and text simultaneously
- Naturalistic movie clips provide dynamic, ecologically valid stimuli
- Automated pipelines eliminate manual inventory development
- Continuous features (semantic distance, completeness) capture gradient impairment

Study aims

1. Evaluate whether main concept (MC) and microlinguistic features from naturalistic movie clip retellings discriminate persons with aphasia (PWA) from healthy controls (HC), hypothesizing large effect sizes for MC features.
2. Determine whether MC features provide incremental validity beyond microlinguistic measures for classification and correlate strongly with aphasia severity.

Study design

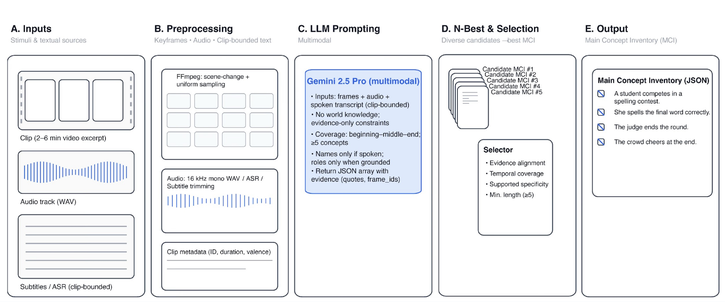
	HC (n = 51)	PWA (n = 54)
Age (years), M ± SD	48.9 ± 16.0	59.5 ± 10.6
Gender (F/M), n	29 / 22	23 / 31
MoCA, M ± SD	28.2 ± 1.3	—
Months post-onset, M ± SD	—	90.5 ± 89.2
WAB-R AQ, M ± SD	—	83.4 ± 16.7
TMT-B/A ratio, M ± SD	—	3.5 ± 2.0
Aphasia type, n (%)	—	Anomic: 38 (70%); Broca's: 5 (9%); Conduction: 3 (6%); Not Aphasic: 6 (11%); TC Sensory: 2 (4%)

Stimuli: 8 movie clips (2–6 min) from DynAMoS database (Girard et al., 2023), selected for emotional variety and narrative richness.

Procedure: Clips presented in randomized order. Participants viewed each clip, then provided a 1-minute verbal retelling. Audio transcribed via CrisperWhisper ASR with manual correction by a licensed SLP.

Features: 20 features: 16 microlinguistic (dptools) + 3 MC features (Gemini 2.5 Pro) + 1 sentiment rating (Gemma 3 12B).

Pipeline: Movie viewing → verbal retelling → CrisperWhisper ASR → feature extraction (microlinguistic via dptools + macrolinguistic via LLM MCA).



Feature derivation

Macrolinguistic (3 MC features via Gemini 2.5 Pro): MC completeness, MC semantic coherence, MC distance to centroid
Sentiment (1 feature via Gemma 3 12B): LLM-based valence rating (1–7 scale)
Microlinguistic (16 features via dptools):

- **Productivity:** word count, MLU morphemes
- **Lexical diversity:** MATTR, Honoré's statistic
- **Lexical frequency:** mean log freq, freq range, high-freq word ratio, mean content word freq
- **Fluency:** total filled pauses, disfluencies per 100 words
- **Morphosyntactic:** clauses per sentence, content word ratio
- **Part-of-speech:** noun, verb, adjective, adverb ratios

Results

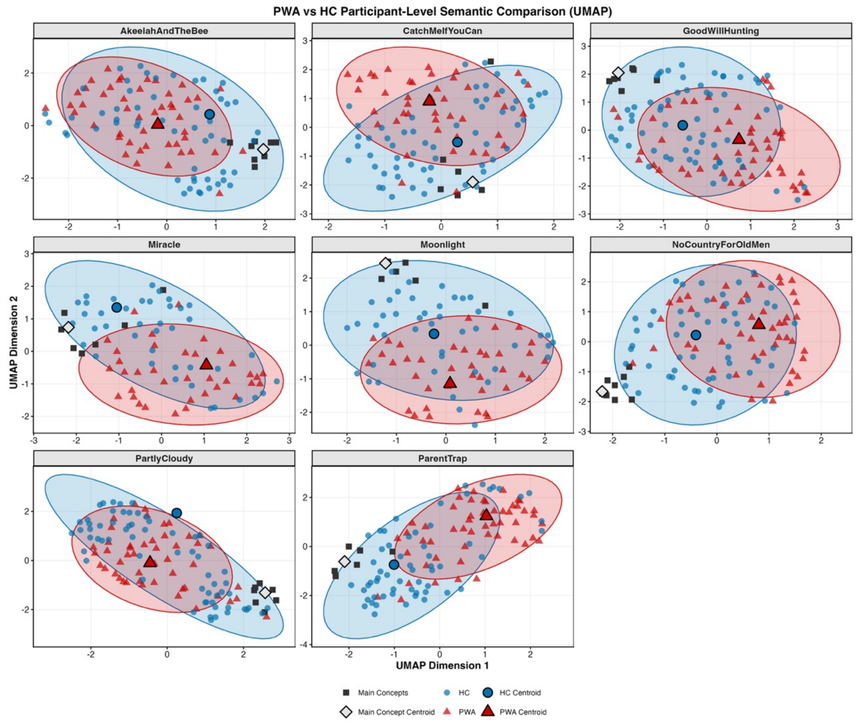


Figure 1. Semantic Space Visualization by Clip. UMAP projection of participant-level semantic embeddings across eight movie clips. Blue circles = HC (n = 51); red triangles = PWA (n = 54); gray squares = main concept positions; large symbols = group centroids and MC centroid. Ellipses show 68% and 95% confidence intervals. HC utterance embeddings cluster tightly near MC centroids, while PWA embeddings show greater dispersion toward peripheral semantic regions (MC distance to centroid: Cohen d = 1.46, p < .001).

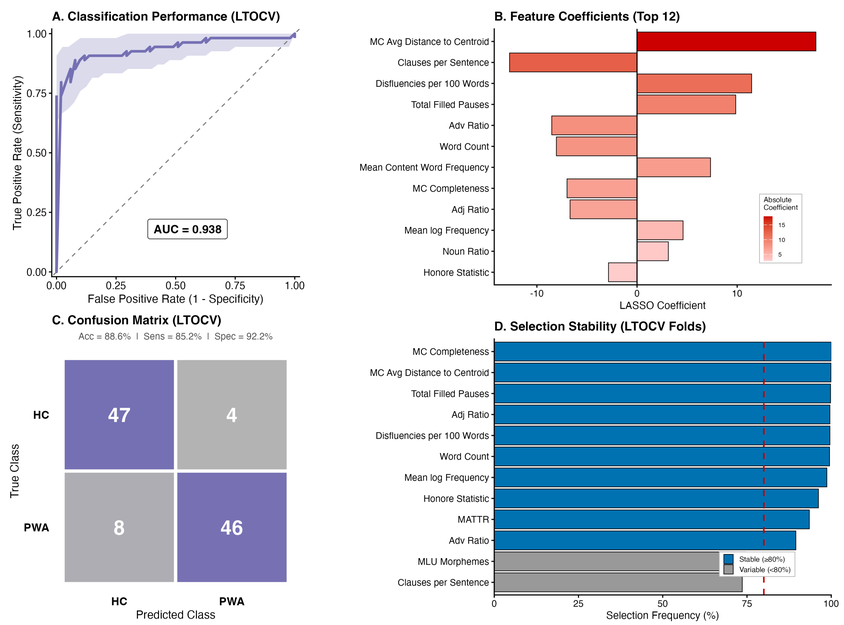


Figure 2. LASSO Classification Performance and Feature Selection. (A) ROC curve from leave-two-out cross-validation (AUC = 0.938). (B) Top 12 LASSO coefficients: MC distance to centroid shows largest positive coefficient (+17.86, elevated in PWA); clauses per sentence shows largest negative coefficient (−12.73, elevated in HC). (C) Confusion matrix: 88.6% accuracy, 85.2% sensitivity, 92.2% specificity. (D) Selection stability across 2,754 LTOCV folds: MC completeness and MC distance to centroid selected in 100% of folds; 9 features with ≥80% stability.

Results (continued)

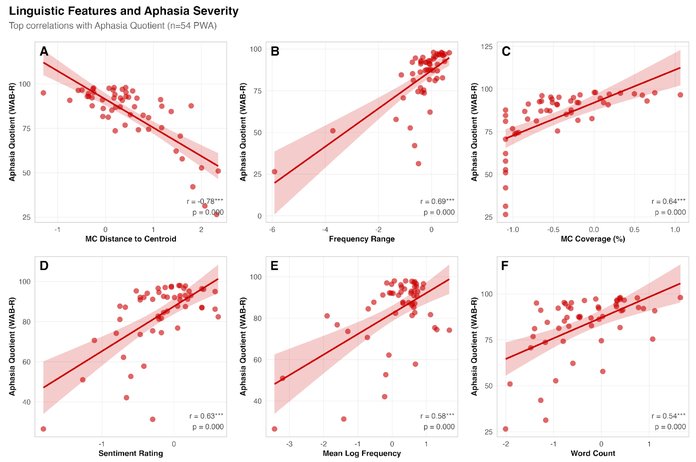


Figure 3. Clinical Correlations with Aphasia Severity. Scatterplots showing relationships between top 6 features (x-axes, z-standardized) and WAB-R Aphasia Quotient (y-axis) within the PWA subsample (n = 54). (A) MC distance to centroid: r = −0.78, p < .001. (B) Frequency range: r = 0.69, p < .001. (C) MC completeness: r = 0.64, p < .001. (D) Sentiment rating: r = 0.63, p < .001. (E) Mean log frequency: r = 0.58, p < .001. (F) Word count: r = 0.54, p < .001. Red lines = linear regression fits; shaded ribbons = 95% CIs.

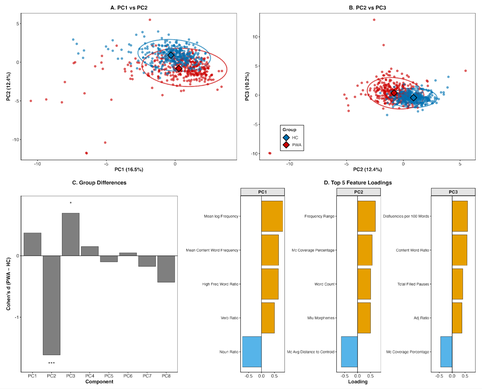


Figure 4. PCA Results. Scatter plots of PWA (red) and HC (blue) across first three principal components with 68% and 95% confidence ellipses, plus feature loadings for PC1–PC3. MC features and microlinguistic measures contribute to distinct variance components.

Conclusion

- Automated multimodal MCA from naturalistic movie clips discriminates PWA from HC with large effect sizes
- MC distance to centroid shows strongest clinical correlation with aphasia severity (r = −0.78)
- MC features provide incremental validity beyond traditional microlinguistic measures (LASSO AUC = 0.938)
- Multimodal LLM pipeline enables scalable MCI generation directly from video stimuli without manual norming

References

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